PersonaScienceBench: Evaluating Scientific Agents with Diverse Simulated Researchers

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Abstract

Scientific agents must adapt to researchers with different expertise and communication preferences, yet benchmark evaluation rarely varies the collaborator. We introduce PersonaScienceBench, an interactive benchmark adapting 200 tasks from five benchmarks across four scientific domains. Persona-conditioned researcher simulators provide context and feedback while task objectives and grading criteria remain fixed across settings; the scientific agent encounters researcher preferences only through interaction. We evaluate seven model configurations using scientific performance, fact recovery, and self-repetition, revealing different strengths across these measures. Detailed analyses with GPT-6 Astra compare four researcher personas with a persona-off control. All ten pairwise comparisons show semantic separation in both simulator and actor final responses (Holm-adjusted $p < 0.05$). Interaction traces show targeted follow-up questions, language accommodation, and workflow revision in response to researcher feedback. These adaptations are not fully reflected in final scores: matched attempts can achieve the same score through different conversations, and some unsuccessful attempts recover every required fact. PersonaScienceBench makes researcher diversity an explicit dimension of scientific-agent evaluation, capturing adaptation during multi-turn scientific work alongside task performance.

1 Introduction

Scientific work is collaborative. Language-model agents now perform substantial parts of the scientific workflow. They search literature, propose hypotheses, write and execute analysis code, and assemble results into reports Lu et al. (2024); Yamada et al. (2025); Gottweis et al. (2025); Mitchener et al. (2025b). The benchmarks used to measure this progress have followed the same trajectory, moving from recall and reasoning questions toward executable analysis tasks drawn from real studies Majumder et al. (2024); Mitchener et al. (2025a); Kon et al. (2025); Ye et al. (2025). Yet in practice these systems function as co-scientists rather than autonomous investigators, and the case has been made that they are not architecturally built for unsupervised discovery Bisht et al. (2026). A deployed scientific agent sits inside a collaboration: someone designed the experiment, ran the instrument, and holds the context that determines whether an analysis is valid. That context is frequently undocumented. Which plates were processed on a different day, why three samples were dropped, whether a covariate is confounded with batch, which normalization the lab actually applies—these facts live with the researcher, not in the data directory. The agent's scientific output therefore depends not only on its own reasoning but on an interaction with a particular human who may explain readily, answer only what was literally asked, or misremember a detail.

The evaluation gap. Most scientific agent benchmarks specify each task completely and uniformly, which measures capability under ideal information but says nothing about capability under realistic information transfer. A recent line of interactive benchmarks closes part of this gap by requiring agents to recognize underspecification and ask: HiL-Bench measures whether coding agents escalate at the right moments Trinh et al. (2026), SciConvBench evaluates multi-turn clarification for ill-posed computational science requests Somasekharan et al. (2026), and work in tool-agent settings evaluates interaction with a simulated user under domain rules Yao et al. (2024); Chen et al. (2026); Ta et al. (2026). These benchmarks establish that asking is hard. They do not, however, vary *whom* the agent is asking. The simulated collaborator is typically a single cooperative interlocutor with one disclosure policy, so the collaborator is a fixed property of the environment rather than an experimental condition. The closest departures from this pattern vary designed behavioral axes—non-collaborative user simulators degrade tool agents through impatience, digression, and

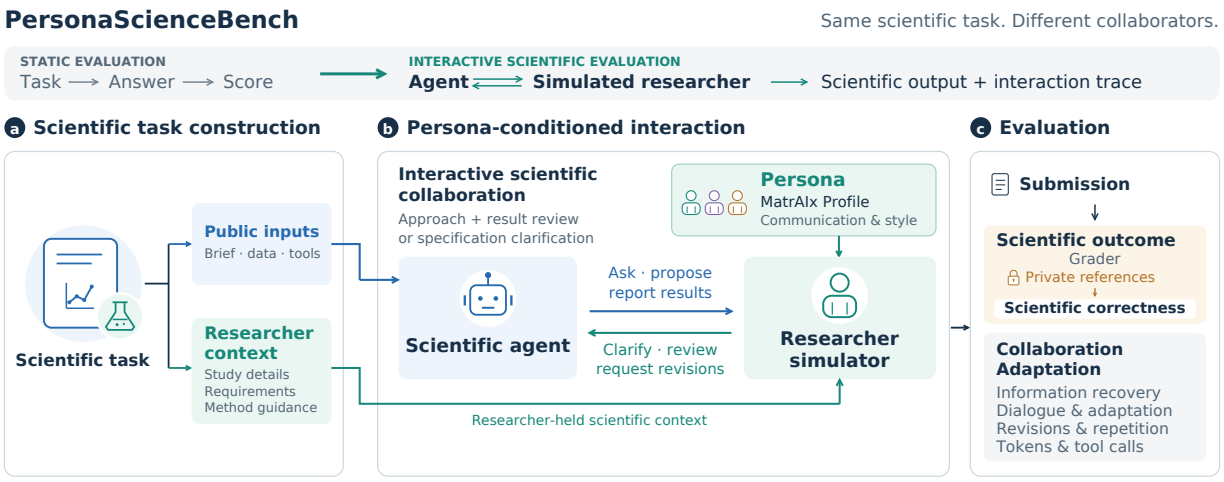


Figure 1: ***Overview of PersonaScienceBench.*** ***a,*** *Scientific tasks are separated into public inputs and researcher-held context.* ***b,*** *A persona-conditioned researcher interacts with the scientific agent through review or clarification.* ***c,*** *Evaluation records scientific outcomes alongside measures of collaboration and adaptation.*

incomplete utterances Shim et al. (2025), and evolved persona policies induce realistic behavioral variation in otherwise cooperative simulators Chopra et al. (2026)—but do so in retail, airline, and customer service domains where there is no scientific ground truth to get wrong. The result is that three distinct sources of difficulty remain entangled in current evaluation: the intrinsic difficulty of the scientific task, the capability of the agent, and the character of the collaborator the agent happens to be working with. Reporting a single score against a single default collaborator silently averages over the third.

Scientific agents therefore need to support multi-turn scientific work in which researchers can clarify requirements, review evidence, and guide revisions. Figure 2 shows that scientific score contributions come from episodes completed over a range of interaction lengths, with different patterns across researcher profiles. Evaluating these interactions alongside final answers captures how scientific work unfolds through collaboration.

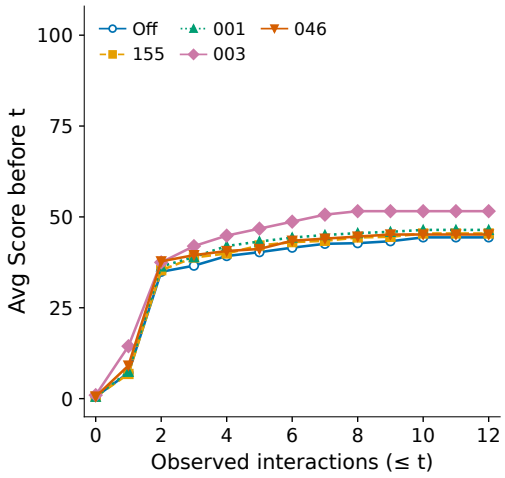


Figure 2: ***Scientific performance across interaction lengths.*** *Cumulative final-score contributions from episodes ending by interaction t across 1,958 runs, divided by all evaluable runs in each setting. Curves summarize observed completion lengths, not within-run learning.*

Our approach. This paper introduces PersonaScienceBench, a benchmark that holds the scientific task fixed and varies the simulated researcher (Figure 1). We convert 200 tasks from five scientific agent benchmarks—DiscoveryBench, BixBench, EXP-Bench, SciConvBench, and ASI-Bench—into collaboration tasks, preserving each source's native deliverable and grading rules rather than reducing them to a common format. An actor agent must produce that native deliverable while interacting with a simulated researcher through a shared review protocol. The researcher is conditioned on a complete rendered profile drawn from MatrAIx, a population-scale simulated-user infrastructure whose persona records span 1,290 categorical dimensions covering background, psychology, capability, and behavior Li et al. (2026). Our renderer emits the populated attributes as an explicit attribute–value listing grouped by category, so a condition is a fully auditable set of stated attributes rather than authored prose; a single profile typically contributes on the order of two hundred such attributes. The profile is injected verbatim into the researcher simulator's system prompt only, without truncation or projection onto the task domain, so the actor observes personality solely through conversation. Every task is also run with the identical protocol and the identity block removed, giving a paired persona-off control that separates collaborator variation from task difficulty. Evaluator references stay private throughout, and each task–condition cell is repeated across independent episodes so that persona effects can be distinguished from ordinary

run-to-run variation. Because the same task is scored under multiple collaborators with a shared baseline, differences are attributable to the collaborator rather than to task selection.

Upstream validation of this persona corpus establishes that conditioning reliably induces declared behavior—91.5% of trials in a controlled adherence study—but does so in Survey, Chatbot, Web, and App environments over a task library concentrated in commerce, software, finance, and healthcare Li et al. (2026). Scientific research collaboration is not among those environments, and product feedback rather than scientific correctness is the measured outcome. PersonaScienceBench supplies that missing environment and asks the question it enables: whether persona-induced behavioral variation propagates into graded scientific error.

What we measure. Evaluation is deliberately two-sided (Figure 1c). Outcome metrics use each source benchmark's grader, so a persona effect registers as a change in scientific correctness and not merely as a change in conversational tone. Interaction metrics—review rounds, revisions, questions asked, clarification coverage, message length, and language adaptation—are recorded separately, because a collaborator can change an agent's behavior substantially without changing its score, and can change its score without changing observable behavior. Keeping these separate is what allows a failure under a difficult collaborator to be diagnosed as a failure to obtain available information rather than assumed to be one.

Contributions. Our contributions are threefold:

- **A benchmark for scientific collaboration with diverse researchers.** We introduce PersonaScienceBench, adapting 200 tasks from five benchmarks across four domains into interactions involving clarification, review, and revision. Researcher personas vary while task objectives and grading criteria remain fixed. Only the simulator receives the profile; the actor encounters researcher preferences through interaction.
- **Evidence of persona diversity and agent adaptation.** We show distinct simulator and actor responses and interaction trajectories across researcher personas. Behavioral analyses and case studies reveal differences in communication volume, targeted follow-up questions, language accommodation, and workflow revision.
- **Evaluation beyond final scores.** We evaluate seven model configurations using scientific performance, fact recovery, and self-repetition. Detailed persona analyses show how different collaboration processes and information-gathering progress can be obscured by final scores, making these behaviors observable alongside task outcomes.

2 Related Work

Language Models with Personas. Conditioning language models on persona descriptions can produce varied responses and simulated user behavior Argyle et al. (2022); Ge et al. (2024). Recent work develops persona generators for user coverage and applies persona-based simulation to interactive-system evaluation Paglieri et al. (2026); Liu et al. (2026). Non-collaborative user simulators and Persona Policies further show that varying user behavior can change tool-agent performance in service domains Shim et al. (2025); Chopra et al. (2026). MatrAIx provides structured profiles spanning background, capability, and behavior Li et al. (2026). We use these profiles to study simulated scientific collaborators, comparing persona-conditioned interactions with independent persona-off repeats to distinguish profile-induced diversity from ordinary run variation.

Interactive Agent Evaluation. Interactive benchmarks increasingly test whether agents ask for missing information and adapt to user feedback. τ-bench evaluates tool-agent-user interaction under domain rules Yao et al. (2024); HiL-Bench targets escalation and help-seeking Trinh et al. (2026); AppWorld-UL and regret-based clarification benchmarks study multi-turn interaction and underspecification Chen et al. (2026); Ta et al. (2026). These settings establish that interaction policy is itself a capability, but they generally hold the collaborator fixed or study non-scientific service domains. PersonaScienceBench instead varies the researcher while retaining a graded scientific objective, allowing collaborator-conditioned behavior and scientific outcome to be analyzed together.

Agents for Scientific Discovery. Language-model agents increasingly support hypothesis generation, experimental implementation, and data analysis, as illustrated by the AI Scientist, AI co-scientist, and Kosmos Lu et al. (2024); Yamada et al. (2025); Gottweis et al. (2025); Mitchener et al. (2025b). Benchmarks evaluate these capabilities through data-driven discovery and computational biology Majumder et al. (2024); Mitchener et al. (2025a), experimental design and execution Kon et al. (2025), and scientific implementation under varying methodological guidance Zhou et al.

(2026). SciConvBench extends evaluation to multi-turn clarification of underspecified computational-science requests Somasekharan et al. (2026). These benchmarks assess scientific task performance and information recovery, but generally leave collaborator behavior fixed. PersonaScienceBench builds on their task settings by varying the simulated researcher on matched scientific tasks and measuring interaction patterns alongside source-specific outcomes. This design tests how collaborator variation affects both the scientific process and its results.

3 PersonaScienceBench

3.1 Persona Construction

Source personas. We draw simulated researcher persona from the public Persona-8B dataset of MatrAIx Li et al. (2026). Its underlying Persona 8B population uses a schema of 1,290 categorical dimensions spanning background, psychology, capability, interaction, and lifestyle. MatrAIx constructs records through two pathways. Synthetic records are sampled in topological order from a directed acyclic dependency graph: source-informed categorical priors and conditional adjustments preserve selected correlations, while compatibility rules rule out specified contradictions. Human-grounded records map biographies, review histories, coded surveys, and consented self-reports into the same schema; fields without source support remain unassigned. Both paths undergo source-appropriate quality checks and deduplication. The released coreset contains 599,847 human-grounded and 400,000 synthetic records. This construction provides varied, inspectable identity prompts; it does not establish that a simulated collaborator represents a particular real researcher or a population subgroup.

From records to simulated researchers. We sampled 300 coreset records, screened them to 180 eligible candidates, and selected 30 profiles before observing any model outcomes. The selection uses source quotas and a greedy coverage objective over directional communication, decision, interaction, tool-collaboration, and workflow attributes. For each selected record, the pinned MatrAIx renderer emits every retained populated attribute as a category-grouped attribute–value identity block; null, placeholder, schema-default, and designated source-dump fields are omitted. We inject this complete rendered block into the researcher simulator's system prompt, without task-specific pruning or summarization. The actor and evaluator never receive it. The persona-off condition uses the same simulator and interaction protocol with the identity block removed. Scientific task facts and grading criteria are held fixed across conditions (Section 3.2).

Table 1 summarizes the backgrounds and interaction preferences of the four profiles used in the detailed persona analyses.

Persona	Background and specialty	Primary language	Interaction preferences
001	Business and management; curriculum design	Hindi	Brief, direct communication.
003	Engineering; electrical work	Russian	Detailed, concrete explanations; speed and novel approaches.
046	Finance and economics; machine learning	Spanish	Formal, indirect communication.
155	Agriculture; agronomy	Mandarin	Detailed guidance; accuracy, familiar methods, and teamwork.

Table 1: Backgrounds and interaction preferences of the four researcher personas.

3.2 From Static Benchmarks to Interactive Simulations

We adapt scientific tasks so that an actor must work with a simulated researcher, exposing decisions that final-answer evaluation alone does not capture. Conversion separates public inputs, researcher context, and private evaluation, and defines when the agents interact to share information (Algorithm 1). The aim is to make consultation and review observable while preserving a well-defined scientific task. Required checkpoints are distinguished from voluntary clarification and revision; longer conversations are not rewarded.

The benchmark contains 200 tasks, with 40 each from DiscoveryBench Majumder et al. (2024), BixBench Mitchener et al. (2025a), EXP-Bench Kon et al. (2025), SciConvBench Somasekharan et al. (2026), and ASI-Bench Zhou et al.

(2026). Table 2 gives their distribution across four domains. The selection covers numerical scientific workflows and diagnostic plots; tasks involving image processing are outside the scope of this study.

Benchmark	Life	Phys./ Eng.	Math./ CS	Soc./ Hum.	Total
DiscoveryBench	5	0	3	32	40
BixBench	40	0	0	0	40
EXP-Bench	0	6	34	0	40
SciConvBench	0	34	6	0	40
ASI-Bench	5	23	12	0	40
Total	**50**	**63**	**55**	**32**	**200**

Domains: Life, life sciences; Phys./Eng., physical sciences and engineering; Math./CS, mathematics and computer science; Soc./Hum., social sciences and humanities.

Table 2: Task counts by benchmark and domain.

The conversion defines what each participant can access and how consultation enters the scientific workflow.

Require: Source task t; persona p (or persona off)
$(b, k, e, \pi) \leftarrow \text{Convert}(t)$
Initialize actor A with public inputs b.
Initialize simulator S with context k and p.
$h \leftarrow []$ {shared conversation}
while protocol π permits another turn **do**
  if π assigns the next turn to A **then**
    $m \leftarrow A.\text{ActAndReport}(h)$
  else
    $m \leftarrow S.\text{Respond}(h)$
  end if
  Append m to h.
end while
$y \leftarrow \text{SealSubmission}(A, h)$
return $e.\text{Evaluate}(y, h)$

Conversion separates public inputs b, researcher context k, and private evaluator e, and selects protocol π. Actor actions use only the tools permitted by the task; evaluator references stay private.

Algorithm 1: Task conversion and interactive simulation.

DiscoveryBench, BixBench, EXP-Bench, and ASI-Bench use approach and result reviews: the researcher approves a proposed approach, then requests an identifiable revision or concludes after reviewing the reported result. SciConvBench retains its native specification-clarification dialogue. Evaluator references remain private, and only the researcher receives the persona profile. Source-specific deliverables and grading are retained within declared adaptations, including EXP-Bench's design-only track and ASI-Bench's objective-only actor with researcher-held methodological guidance. Appendix A.1 details the shared conversion rules, information boundaries, interaction protocol, and acceptance rubric; Appendices A.2–A.6 describe the benchmark-specific implementations.

4 Experimental Setup

4.1 Models, Persona Settings, and Evaluation Protocol

We evaluate Claude Opus 5.5 Anthropic (2026) and Claude Haiku 4.5 Anthropic (2025) from Anthropic, and Gemini 3.7 Flash Google DeepMind (2026) and Gemini 3 Flash Google DeepMind (2025) from Google DeepMind. The OpenAI models are GPT-6 Astra OpenAI (2026), GPT-5 mini, and GPT-5 nano OpenAI (2025) (Table 3). Each configuration uses the same model for the actor and researcher simulator; model-based judging uses GPT-6 Astra. Astra runs use low reasoning effort. The detailed persona analyses in Sections 5.2–5.5 use the set of runs with GPT-6 Astra.

We use four contrasting persona profiles to examine how agents adapt to researchers with different disciplinary backgrounds and collaboration preferences (Table 1). Persona 001 combines a business and management background with a curriculum-design specialty and a preference for brief, direct communication. Persona 003 brings an engineering background and favors detailed, concrete explanations, speed, and novel approaches. Persona 046 combines finance and economics with a machine-learning specialty and formal, indirect communication. Persona 155 brings an agriculture and agronomy background, emphasizing detailed guidance, accuracy, familiar methods, and teamwork. Together, these profiles provide contrasting contexts for examining how agents explain technical work, elicit missing information, and negotiate methodological choices. Only the researcher simulator receives the profile; the actor encounters its expression through interaction. Persona off uses the same task and interaction protocol with the profile omitted.

The current design uses one trial per task and persona setting for each model. Analyses use evaluable task attempts, with coverage determined separately for each metric. When repeated historical runs are included, they are averaged within each task. The domain table includes valid repaired outputs where available. Astra scorecard estimates use original runs, with 95% confidence intervals from 20,000 study-cluster bootstrap resamples within benchmarks Field and Welsh (2007). Resampling draws are shared across persona settings. These intervals describe variation across sampled studies, conditional on the recorded runs.

4.2 Evaluation Metrics

Scientific performance. We normalize each benchmark's score to a 0–100 scale. Performance for setting p is $S_p = |\mathcal{B}|^{-1} \sum_{b \in \mathcal{B}} |Q_b|^{-1} \sum_{q \in Q_b} s_{qp}$, where s_{qp} is the normalized question score, Q_b contains questions observed under all five settings in benchmark b, and $\mathcal{B}$ contains applicable benchmarks. Mean Persona Performance is $S_{\text{persona}} = |\mathcal{P}|^{-1} \sum_{p \in \mathcal{P}} S_p$ for $\mathcal{P} = \{155, 001, 003, 046\}$; persona off is reported separately.

Fact recovery. For SciConvBench disambiguation cases, fact recovery is $F_e = 100\,|H_e \cap R_e|/|H_e|$, where H_e is the nonempty set of required withheld facts and R_e is the set recovered during episode e.

Actor self-repetition. We measure lexical reuse across actor messages using word windows. For window w in eligible message m, the match indicator is $I_{em}(w) = \mathbf{1}\{\max_{v \in W_{e,<m}} \sum_{k=1}^{6} \mathbf{1}\{w_k = v_k\} \geq 3\}$, where $W_{e,<m}$ contains windows from earlier eligible actor messages and $\mathbf{1}$ denotes the indicator function. Episode-level repetition is $\text{Rep}_e = \frac{100}{M_e - 1} \sum_{m=2}^{M_e} |W_{em}|^{-1} \sum_{w \in W_{em}} I_{em}(w)$, where W_{em} contains the windows in message m and $M_e \geq 2$ is the number of eligible messages.

5 Results

5.1 Model-Level Scientific Collaboration

Actor model	Persona-off performance	Mean Persona Performance	Fact recovery (%)	Self-repetition (%)
Anthropic Claude				
Claude Opus 5.5	49.48 [45.02, 54.14]	50.55 [48.15, 53.06]	98.95 [97.83, 99.86]	16.23 [14.79, 17.72]
Claude Haiku 4.5	21.97 [16.25, 27.71]	19.72 [15.19, 24.14]	67.95 [61.77, 74.37]	21.43 [17.23, 24.53]
Google Gemini				
Gemini 3.7 Flash	48.21 [42.73, 53.77]	48.17 [45.56, 50.83]	75.24 [67.27, 82.79]	13.08 [11.05, 15.13]
Gemini 3 Flash	33.07 [28.55, 37.91]	31.98 [29.46, 34.42]	90.01 [85.65, 93.85]	13.43 [11.72, 15.23]
OpenAI GPT				
GPT-6 Astra	47.54 [39.11, 55.46]	48.82 [40.46, 56.67]	88.86 [76.96, 97.68]	12.69 [11.65, 13.73]
GPT-5 mini	26.37 [22.19, 30.94]	31.63 [28.32, 35.10]	68.42 [60.71, 75.99]	10.52 [8.78, 12.33]
GPT-5 nano	14.56 [10.41, 18.69]	15.55 [13.29, 17.83]	63.26 [55.83, 70.76]	15.99 [13.70, 18.44]

Table 3: ***Model-level scientific collaboration scorecard.*** *Means with 95% confidence intervals. Haiku fact recovery and self-repetition, including their 95% confidence intervals, are collaborator-reported; the interval computation method has not been independently verified. Performance uses scores normalized to 0–100; persona means weight the four profiles equally. GPT-6 Astra uses low reasoning effort. Each actor also serves as its researcher simulator; model-based judging uses GPT-6 Astra.*

Across the seven reported model configurations, the difference between mean persona performance and persona-off performance varied in both direction and magnitude (Table 3). GPT-5 mini showed the largest observed increase, from 26.37 under persona off to 31.63 across the four personas, a difference of 5.26 points. Gemini 3.7 Flash had nearly identical means under the two settings (48.21 and 48.17, respectively). Evaluating multiple researcher settings therefore reveals variation in scientific performance that a single persona-off score cannot capture.

The model configurations also showed contrasting strengths in task performance and interaction behavior. Gemini 3.7 Flash had higher mean persona performance than Gemini 3 Flash (48.17 versus 31.98), whereas Gemini 3 Flash had higher fact recovery (90.01% versus 75.24%). GPT-5 mini had the lowest observed self-repetition (10.52%), further distinguishing interaction behavior from the ordering by performance. Together, these measures characterize task success, recovery of missing information, and reuse of earlier language, providing a fuller account of how agents participate in scientific collaboration. The following analyses examine how individual researcher profiles shape these behaviors.

5.2 Performance across Scientific Domains

Domain	Off	155	001	003	046
Life	27.61 [9.41, 46.54]	28.12 [9.85, 47.09]	27.24 [9.07, 46.10]	28.23 [7.29, 49.68]	27.02 [8.50, 47.30]
Phys./Eng.	66.73 [58.70, 74.25]	67.25 [59.68, 74.42]	69.71 [62.31, 76.54]	76.91 [64.40, 87.98]	66.34 [55.83, 76.16]
Math./CS	50.67 [43.72, 57.35]	52.21 [45.38, 58.65]	54.15 [47.30, 60.71]	57.29 [54.67, 59.50]	50.76 [43.61, 58.19]
Soc./Hum.	14.97 [8.33, 18.33]	17.80 [12.10, 24.89]	16.86 [9.98, 26.52]	19.60 [13.47, 27.30]	18.35 [13.00, 24.17]
Avg.	39.99 [34.35, 45.34]	41.34 [35.70, 46.92]	41.99 [36.29, 47.84]	45.51 [38.90, 51.81]	40.62 [34.54, 46.69]

Domains: Life sciences; physical sciences and engineering; mathematics and computer science; social sciences and humanities.

Table 4: ***Performance by scientific domain.*** *Mean normalized score (0–100), with pointwise 95% confidence intervals below. Columns identify simulator personas; Avg. weights domains equally.*

The persona profiles showed different relative strengths across scientific domains (Table 4). Although persona 003 had the highest observed mean throughout, the ordering of the other profiles changed with the domain. Persona 001 ranked second in physical sciences and engineering and in mathematics and computer science, whereas persona 046 ranked second in social sciences and humanities. Their relative performance reversed: 001 scored above 046 in mathematics and computer science (54.15 versus 50.76), but below it in social sciences and humanities (16.86 versus 18.35). Thus, an overall persona ranking can obscure domain-specific strengths.

The profiles provide useful context for these differences. Persona 001 combines business and management with curriculum design and brief, direct communication; persona 046 combines finance and economics with formal, indirect communication. Their contrasting domain patterns suggest that researcher background and interaction preferences are relevant together when examining agent specialization. Persona 155, whose agronomy background accompanies detailed, accuracy-oriented guidance, ranked second in life sciences, although means were closely grouped in that domain. Persona 003's engineering background and concrete, analytical style accompanied its largest difference from persona off in physical sciences and engineering (76.91 versus 66.73). Together, these patterns motivate evaluating agents across combinations of researcher profiles and scientific domains to reveal strengths that an aggregate score can conceal. Domain estimates use each setting's observed task coverage.

5.3 Persona Diversity Shapes Agent Responses and Interaction Paths

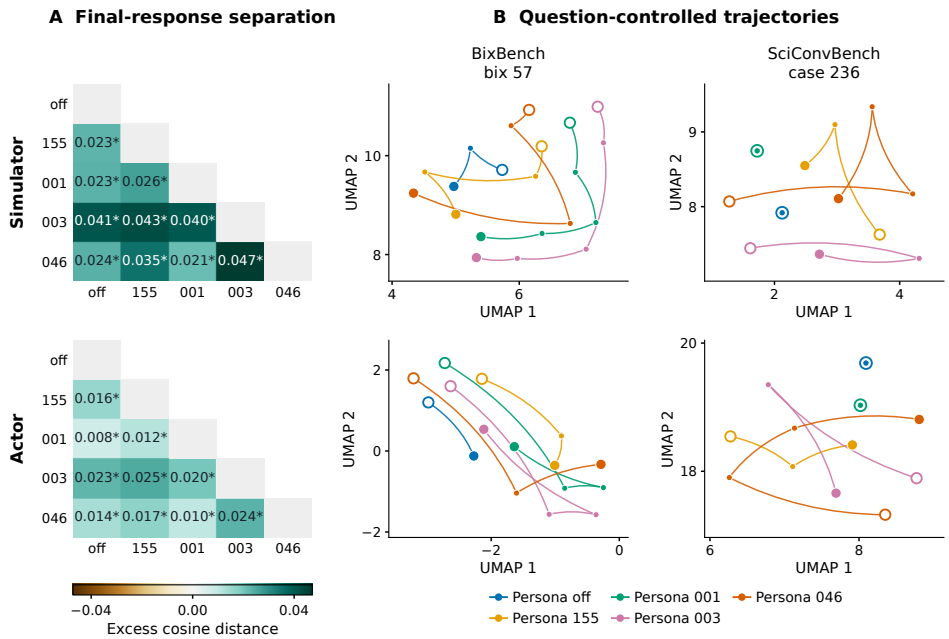


Figure 3: ***Semantic separation and interaction trajectories. A,*** *Final-response semantic separation for simulators (top) and actors (bottom); stars indicate Holm-adjusted $p < 0.05$.* ***B,*** *Example UMAP trajectories for BixBench and SciConvBench. Hollow circles mark starts, filled circles mark ends, and small dots mark intermediate messages; overlapping start/end markers denote a single message.*

Researcher personas were associated with distinct final responses from both the simulator and the actor (Figure 3A). We compared Qwen3-Embedding-0.6B Zhang et al. (2025) embeddings within each domain, measured separation relative to within-persona variation across questions, and weighted the four domains equally. All ten persona-pair comparisons showed positive separation in each role (Holm-adjusted $p < 0.05$ in question-wise persona-label permutation tests Phipson and Smyth (2010); Holm (1979)). The effect of personas at the same time extended to the actor's final responses, even though the actor encountered the researcher profile only through conversation.

Persona diversity appeared in both the intermediate messages and the overall course of interaction (Figure 3B). In BixBench, both simulator and actor trajectories followed distinct paths with similar overall movement and partial convergence in the UMAP McInnes et al. (2018) projection. The shared task thus elicited a recognizable progression across personas, while the messages produced along the way remained different. In SciConvBench, the simulator trajectories differed in their overall patterns, and the corresponding actor trajectories were distinct as well. These examples illustrate how diverse researcher behavior was reflected in the agent's responses throughout the interaction.

Persona-conditioned interactions also drew the actor into more proactive participation in multi-turn scientific work. The BixBench example involved more interaction iterations under every persona than under persona off. In SciConvBench, personas 155, 003, and 046 involved three or four interaction iterations, compared with one iteration for persona off. During these conversations, the actor pursued specific questions: under persona 046, it successively clarified the starting structure, magnetic configuration, external pressure, and convergence criteria. This progression from a brief interaction to targeted follow-up questions makes the agent an active participant in specifying how the task should proceed. Such participation is valuable in scientific collaboration, where initial requests can leave consequential assumptions and preferences implicit. Continued dialogue gives the researcher and agent opportunities to make those choices explicit, resolve uncertainty, and incorporate feedback before committing to an approach.

5.4 Researcher Profiles Elicit Different Communication Patterns

Profiles that favor detailed communication were accompanied by more extensive output from both the simulator and the actor (Figure 4A,B). Relative to persona off, personas 155 and 003 elicited an additional 1.12 and 0.61 thousand actor tokens per episode, respectively. Simulator output also increased under both profiles; all four comparisons had Holm-adjusted $p < 0.05$. These comparisons matched questions and trials before averaging across benchmarks. The contrast also appeared between personas: actor output under 155 and 003 exceeded that under persona 001, whose profile favors brief, direct communication. Researcher preferences were thus reflected in how much the agent produced in response, extending the behavioral variation beyond the simulator itself.

The clearest differences appeared in communication volume. Clarification recall varied by −0.28 to 4.55 percentage points across persona pairs, and tool use by −0.54 to 0.28 calls per episode (Figure 4C,D); neither metric showed a pairwise difference at Holm-adjusted $p < 0.05$. These patterns give the benchmark contrasting communication demands: detailed, accuracy-oriented guidance under persona 155, concrete explanations under persona 003, and concise responses under persona 001. Evaluating these settings together makes responsiveness to researcher preferences an observable part of scientific collaboration.

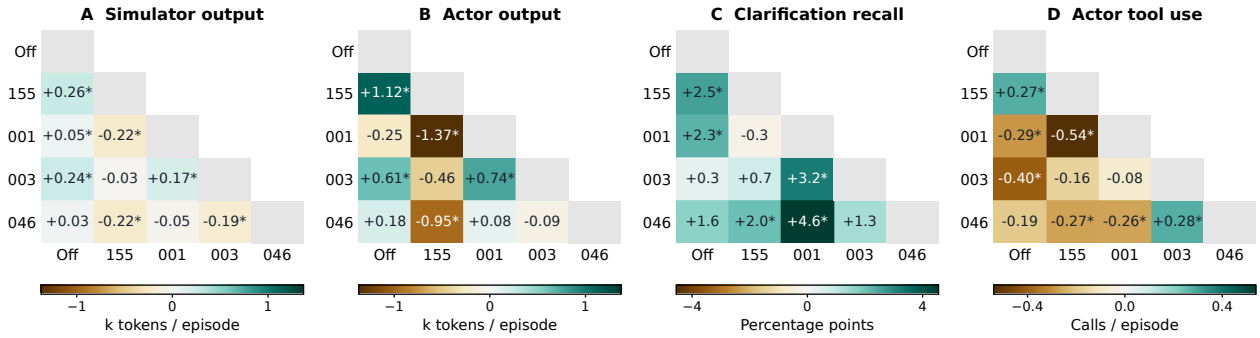


Figure 4: ***Pairwise behavioral differences.*** *Cells show row minus column.* ***A, B,*** *Simulator and actor output (thousands of tokens per episode).* ***C,*** *SciConvBench clarification recall (percentage points).* ***D,*** *Actor tool calls per episode. Stars:* ***A, B,*** *Holm-adjusted $p < 0.20$;* ***C, D,*** *absolute differences above 2 percentage points and 0.2 calls, respectively.*

5.5 Interaction Behavior beyond Final Scores

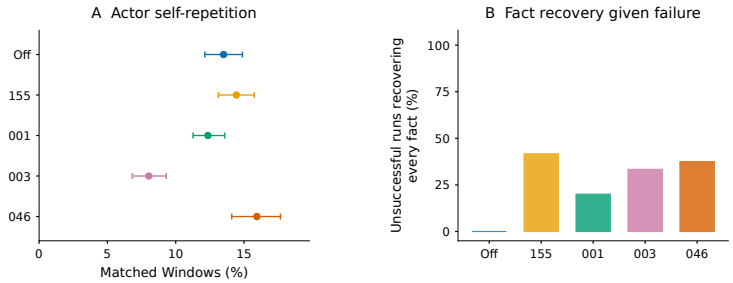


Figure 5: ***Interaction metrics across researcher personas.*** ***A,*** *Actor self-repetition; bars show pointwise 95% study-cluster confidence intervals.* ***B,*** *Complete-fact recovery among failed SciConvBench disambiguation episodes.*

The actor returned to its earlier wording to different degrees depending on the researcher persona (Figure 5A). Self-repetition was 8.03% under persona 003, compared with 13.50% for persona off and 15.94% under persona 046. These differences are relevant to conversational mode collapse, where interactions become concentrated in a few recurring response patterns. CORE evaluates this phenomenon jointly through mode diversity, lexical repetition, and semantic stagnation Pandey et al. (2026). Extended closed-loop LLM studies also report semantic collapse, with meanings converging even as surface wording varies Kong et al. (2026). This motivates assessing self-repetition alongside semantic diversity (Section 5.3). In scientific collaboration, avoiding unnecessary repetition preserves space for targeted clarification, updated interpretations, and revision of the scientific approach. The lower self-repetition under persona 003 highlights a setting in which less of the actor's output reuses its earlier language. The observed interaction lengths further motivate evaluating multi-turn scientific work (Figure 2). Every setting received score contributions from episodes completing after the second iteration. For persona 003, the cumulative contribution rose from 37.49 at iteration two to 48.69 at iteration six as later-completing episodes were included.

Information recovery revealed progress during multi-turn scientific work, including in episodes that ultimately failed (Figure 5B). Within each setting's failed SciConvBench disambiguation episodes, the case-averaged rate of recovering all required facts ranged from 20.0% to 41.7% across the four personas, compared with 0% for persona off. The persona-conditioned episodes with complete recovery contained four to ten interaction iterations, during which the actor elicited every annotated missing fact before termination. This identifies a concrete intermediate accomplishment within unsuccessful tasks: establishing the full factual context needed to formulate a scientific response. Final scores alone would leave this information-gathering progress unreported.

6 Discussion

Interactions with persona 155 illustrate how agents adapt their methods and communication during multi-turn scientific work (Figure 6, Appendix B.1). In BixBench, researcher feedback prompted the actor to replace an unavailable R/limma Ritchie et al. (2015) workflow with Python analysis while retaining the scientific objective. In SciConvBench, the actor switched from English to Chinese and clarified requirements through five interaction iterations, compared with one under persona off. Both SciConvBench attempts received full credit, yet followed different communication and clarification paths. Together, these cases show how agents can respond to execution constraints, accommodate researcher preferences, and refine their understanding of the task. Persona-based evaluation makes these aspects of scientific collaboration visible alongside final scores, capturing how an agent works with a particular researcher to pursue the task. Full interactions appear in Appendices C.1, C.2, and C.3.

A. Revising an analysis (C2)
BixBench; persona 155; translated from Chinese.
Actor, event 28: […] the approved limma-trend main analysis is blocked by the environment […]
Researcher, event 29: […] There is no need to try installing R again: limma was our proposed approach rather than a fixed task requirement; […] you may explicitly switch to the available Python tools to fit ordinary least squares on log2(count+1) […]
Actor, event 36, evidence: Execution succeeded (exit code 0): `analysis/ols_analysis.py`. […] All three contrasts had 17,049 valid gene tests. […]
Full interaction: Appendix C.1.

B. Accommodating the researcher (C1)
SciConvBench; persona 155.
Actor, event 2, English: What torque acts on the shaft (specify its magnitude, or indicate that it should remain a symbolic parameter)?
Researcher, event 3, Chinese, translated: The torque applied to the shaft is $T = 1{,}000{,}000\,\mathrm{N{\cdot}mm}$. Please use this value in the calculation; there is no need to retain it as a symbolic parameter.
Actor, event 4, Chinese, translated: What is the shaft's torsional length (in mm)?
The dialogue continues in Chinese, with five clarification interactions in total; the matched no-persona interaction uses one.
Full interactions: Appendices C.2 and C.3.

*Figure 6: **Adaptation during scientific collaboration.** Selected interactions show workflow revision (A) and language accommodation (B). Highlights are added for emphasis; […] marks omitted text. Chinese excerpts are translated into English. Event numbers retain their source identifiers.*

7 Conclusion

We introduced PersonaScienceBench, which adapts 200 tasks from five scientific benchmarks for collaboration with diverse simulated researchers. Scientific objectives and grading criteria remain fixed across persona settings, allowing researcher variation to be examined within the same task. Persona-conditioned interactions produced distinct agent responses, communication patterns, and interaction trajectories. Agents asked targeted follow-up questions, accommodated language preferences, and revised analysis workflows in response to researcher feedback. Final scores alone did not capture these differences in how the work was carried out. By evaluating interaction behavior alongside scientific performance, PersonaScienceBench makes responsiveness to different researchers an explicit part of assessing agents for multi-turn scientific work.

8 Limitations

PersonaScienceBench covers 200 computational tasks drawn from five source benchmarks, with an emphasis on reasoning over text, data analysis, and code execution. This scope leaves analysis of images and workflows involving physical experiments for future extensions. Including these settings would allow evaluation of how agents respond to visual evidence, instrument constraints, and researcher feedback during experimental work. The current protocols also organize collaboration into bounded episodes between one actor and one simulated researcher. Longer projects and interactions with multiple researchers would introduce additional demands, including retaining context across sessions, revisiting earlier decisions, and coordinating differing priorities. Extending the benchmark along these dimensions would test whether the observed adaptations persist in multi-turn scientific work across broader workflows and longer projects.

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A Details on Benchmark Implementation

A.1 General Principles and Conversion Rubric

This subsection expands the conversion and interaction contract summarized in Algorithm 1. The shared rules specify what an adapter must preserve, what a researcher may know, and when interaction can change the actor's next action. The following benchmark subsections are reserved for their specific implementations.

Motivation: make collaboration consequential and observable. A solve-and-submit interface can measure the quality of a final scientific artifact without exposing how its producer negotiates assumptions, obtains context, or responds to criticism. Adding an optional question channel alone also permits an episode to finish without consulting the researcher. Our adaptation therefore gives the researcher an explicit role in reviewing proposed approaches and reported results. These checkpoints create opportunities to clarify a consequential choice, request evidence, or identify a revision before submission. The intended improvement is coverage of collaborative scientific work, rather than a guaranteed increase in accuracy or conversation length. A required handoff is a protocol event, not evidence of spontaneous actor initiative. Additional questions and revisions must be measured separately. SciConvBench already evaluates clarification, so it retains its native dialogue instead of receiving artificial approval interactions.

Preserve scientific identity and solvability. Each converted task retains its native identifier, source revision, study group, scientific objective, required deliverables, and an explicit evaluation contract. Fixed requirements, including units, methods, output schemas, thresholds, and data splits, remain unchanged unless the conversion is declared as a distinct variant. The public brief must expose the basic scope, permitted resources, access instructions, and submission format. Researcher-held requirements must be discoverable within the declared interaction budget. We do not remove arbitrary information or invent missing facts merely to induce questions. A useful researcher role can instead consist of discussing choices or reviewing evidence already accessible to the actor. Shared input data do not by themselves justify merging distinct questions into one episode.

Allocate information by role. Conversion produces separate public inputs b, researcher context k, and evaluator configuration e, together with interaction protocol π. The actor receives the public brief, allowlisted files and tools, actual runtime information, and shared researcher replies. The researcher receives permitted task context, its persona, and the shared conversation, including actor-selected evidence. It does not automatically inspect private actor reasoning, tool logs, or unshared files. Reference answers, gold solution artifacts, and hidden tests remain evaluator-only; the controller manages versions, access, and records. Exporters select explicit fields and files rather than passing through entire source records or repositories. Access boundaries must hold in mounts and tool interfaces, not merely in filenames. Native labels or metadata that are legitimate task inputs may remain public. Source-context access is distinguished from access to an answer key.

Ground context and distinguish advice from requirements. Each extracted fact has a stable identifier, a source revision and path, and an inspectable locator, such as a field or text span. Facts are typed as context, fixed requirements, or optional advice; unsupported details remain unknown. An empty fact pack is valid when the source supplies no eligible additional context. Automated extraction and validation are recorded as automated processing rather than human review. Reference material may be inspected during curation to detect leakage, but it is not used to synthesize the researcher's runtime knowledge. The full-profile study permits general scientific knowledge and suggestions in both persona conditions. Such suggestions do not become new task requirements, and source-specific assertions still require grounding. A valid fact identifier does not independently verify every sentence of a generated reply. Claims about available files, packages, or hardware must match the actual runtime.

Separate researcher identity from the scientific task. The persona is a versioned configuration distinct from the task context. Persona off retains the same common instructions, knowledge permissions, and interaction protocol with the identity block removed. The actor and evaluator do not receive that block. Across persona conditions, fixed task facts, objectives, tools, budgets, and evaluation settings are intended to remain constant; any departures must be recorded. Actual questions, disclosures, and review decisions may differ as the conversation develops. A change to knowledge access or disclosure policy is a separate experimental factor, not solely a change in communication style. Persona

instructions cannot authorize fabricated evidence or override scientific constraints. These rules define the comparison; they do not establish that historical runs with unequal coverage or evaluator versions achieve perfect control.

Require meaningful checkpoints, not extra dialogue. For execution and design tracks, the researcher first requests inspection of the inputs and a proposed approach. The actor's plan handoff reports its observations, choices, evidence, assumptions, and uncertainties. The researcher approves the approach or requests a focused clarification. After approval, the actor performs the analysis or prepares the design and returns a result handoff with the required deliverable and supporting evidence. The researcher may request a revision for an identifiable mismatch or unresolved uncertainty, or conclude the episode. It is instructed to distinguish actor reports from independently verified findings and never to act as the final grader. There is no minimum quota of additional questions and no reward for extending the conversation.

The actor's workspace and both roles' conversation histories persist within an episode and reset between independent episodes. Tool calls, researcher questions, shared progress, and review decisions are recorded separately. Protocol and resource limits are declared per run. In the recorded `researcher-review-v2` protocol, approach approval is required before a result handoff, and researcher-selected stopping is allowed only during result review. Thus an unresolved input problem can consume the budget in approach review without producing a submission; this behavior must remain visible in historical interpretation. A revised stopping rule defines a new protocol version. SciConvBench instead ends on a completed specification or its native dialogue limit and does not execute a scientific solver.

Seal deliverables and retain source-specific evaluation. The controller checks the submission contract and seals the selected final artifact before evaluation. A plan, an incomplete handoff, or a budget exhaustion is not silently promoted to a successful final submission. In the review protocol, a complete candidate is submitted only after the researcher selects termination. Public format checks and normal execution feedback are permitted, but hidden evaluator feedback cannot become an iterative hint channel. Researcher approval is therefore distinct from scientific correctness. Execution-based graders remain execution-based; LLM-scored tracks record their judge, prompt, parser, and aggregation versions. Converting scores to a common numerical scale does not make their scientific meanings equivalent.

Declared adaptations are part of this contract. EXP-Bench evaluates proposed experimental designs rather than executed experiments. ASI-Bench gives the actor its objective-only request and the researcher eligible guidance from the next-richer prompt level, creating a collaborative variant of the original condition. BixBench schema context can also be found in public inputs, whereas SciConvBench's intended specification supplies researcher- held context. These differences are retained rather than forcing identical information asymmetry or deliverable types across benchmarks.

Acceptance rubric and traceability. Table 5 defines the evidence required to accept a conversion. It is a construction rubric, not an additional score assigned to an actor or conversation. Adapters produce a versioned task package or a recorded exclusion without per-case manual editing. Catalog conversion, asset readiness, successful execution, and scientific correctness are separate states. A schema-valid package is not evidence that the source context is scientifically adequate or that the runtime is ready.

Criterion	Required evidence
Scientific identity	Native task and study identifiers, objective, fixed constraints, deliverables, and a record of intentional changes.
Useful interaction	A legitimate context, decision, or review role for the researcher; no invented facts or quota of extra interactions.
Information separation	Allowlisted actor/researcher exports and runtime access checks that exclude hidden answers, tests, and solution artifacts.
Grounding and persona	Fact provenance, explicit unknowns and advice, versioned persona rendering, and declared knowledge/disclosure policies.
Runtime readiness	Resolved input roles and assets, hashes and mount mappings, actual resources, permitted tools, and required output formats.
Evaluation compatibility	Native submission mapping, evaluator version, declared design/execution scope, scoring changes, and error behavior.
Outcome accounting	Attempted, submitted, evaluated, and scored counts; distinct scientific and infrastructure failures; explicit retries and missing outcomes.
Reproducibility	Source, conversion, context, persona, runtime, and evaluator versions, with trace and submission identifiers and source attribution.

Table 5: Acceptance rubric for benchmark conversion. Each criterion requires inspectable evidence; unresolved checks remain explicit rather than being inferred from successful serialization.

All attempts remain in the record, including missing submissions, timeouts, and provider or evaluator failures. Any penalty or repair policy must be stated while preserving the original status; a retry is not an independent trial or permission to select the best result. Related questions retain study identities for analysis. Automated checks target concrete failures, such as leaked answer fields, unknown input roles, invalid artifact mapping, or missing outcomes disappearing from an aggregate. The rubric specifies what must be checked; it does not assert that every source case has passed all checks or is runnable in every environment.

A.2 DiscoveryBench

DiscoveryBench Majumder et al. (2024) is converted into a collaborative data-analysis task whose deliverable remains a natural-language hypothesis supported by the supplied data. The actor receives the original question, dataset files, and public dataset and column descriptions. The simulator receives source-provided domain knowledge, separated into provenance-linked context or advice, but receives neither the reference hypothesis nor the reference workflow. Missing study details remain unspecified. Questions from the same study retain their study identifier, even when they use different metadata records. For test tasks, evaluation references are retrieved from the released answer key rather than the placeholder hypothesis fields in task metadata.

The shared review protocol makes the proposed analysis and its reported evidence available for researcher feedback: the actor first presents its approach, then analyzes the data and submits its hypothesis for review. An analysis workflow and supporting code or outputs may accompany the final hypothesis. Scoring retains the decomposition into scientific context, variables, and relations: context recall is multiplied by the mean product of variable F1 and relation agreement. The configured judge is connected to the evaluator, while reference material remains evaluator-only. This score measures agreement between the submitted and reference hypotheses; it does not independently verify that an accompanying analysis was executed correctly. The conversion therefore records shared evidence and supporting artifacts separately from hypothesis agreement.

A.3 BixBench

BixBench Mitchener et al. (2025a) retains the original open-ended question and its associated data capsule, including any specified variables, methods, and adjustments. Conversion exports permitted input files from the capsule and excludes reference notebooks and answer-bearing annotation fields. Archive and extracted-file hashes identify the inputs used. Unresolved file roles prevent a partial input inventory from being declared ready. Researcher context is derived from those inputs, including available column names, worksheet structure, identifiers, labels, and file-format information. Automatically extracted schema facts are distinguished from reviewed annotations. This context is also accessible through the actor's own data inspection; interaction provides an opportunity to explain or review the analysis rather than

an artificially exclusive route to the schema. Unspecified code meanings and preprocessing choices are not supplied as facts.

After approach approval, the actor performs the analysis and presents its answer and evidence for researcher review. A final submission requires both an explicit declaration that the analysis was executed and an analysis script or notebook. These requirements form an additional submission gate; they do not establish that the analysis is scientifically correct. The answer is then evaluated with the task's string, range, or language-model verifier against the private ideal answer. The original verifier selection and answer-parsing rules are retained. Thus, the benchmark score concerns answer correctness, while the analysis artifact and interaction trace support separate examination of how that answer was obtained.

A.4 EXP-Bench

EXP-Bench Kon et al. (2025) is adapted as an explicitly design-only task. The actor receives the scientific question and approved public task material in a staged workspace, and develops an experimental design with a rationale, controls, an analysis plan, and feasibility considerations. Researcher context is selected from source setup passages using cached, automated classification. The classifier selects exact source spans as background or fixed requirements; a requirement is eligible only when established by the question. Expected outcomes, procedural recipes, and experimental choices that the actor is supposed to make are excluded. The classifier never receives the reference design. Empty context sets are permitted, and cases with invalid references or mismatched questions and references are excluded during construction.

The researcher reviews the proposed approach before the actor develops the full design, then requests revisions or accepts the completed design. The accepted text is saved as the design checkpoint; neither experimental execution nor an empirical conclusion is required. Evaluation uses the released reference-coverage judge, comparing the design with private reference design elements. Repeated judgments are retained and averaged, and the released integrity monitor is applied separately to the saved trace. Both the coverage score and the monitor-gated score are recorded: a rejected trace receives zero after gating, while transport failures remain evaluation failures. Invalid monitor output is retried under the released parsing policy and receives a distinct parse-error status if unresolved. These outputs measure reference coverage under the design-only adaptation, not successful execution or reproduction of the source experiment.

A.5 SciConvBench

SciConvBench Somasekharan et al. (2026) already makes clarification central to the task, so conversion preserves its native dialogue rather than adding approach and result reviews. The actor receives either an incomplete scientific request or a request containing inconsistencies. The simulator holds the corresponding complete or corrected specification and answers clarification questions under the assigned persona. This specification represents the researcher's intended requirements, not a solution produced by a scientific solver. Native issue annotations are retained for evaluation. Construction requires a valid request, intended specification, and issue annotations, preserves source case identities, and applies the native 20,000-character request limit without filtering on previous models' success.

The actor can ask up to ten clarification questions and can terminate earlier by marking its specification complete. If the question limit is reached, the runner requests a final specification. The specification, conversation, and intent judges evaluate whether missing information or inconsistencies were resolved, whether the relevant issues were addressed in dialogue, and whether the final specification captures the intended task. The implementation retains recovery before the intent gate as well as the gated outcome, and separately records clarification precision, clarification recall, and memory consistency. Keeping specification recovery and conversational grounding separate distinguishes resolution supported by dialogue from silent resolution. No scientific solver is executed, so success here concerns recovering and expressing the intended task rather than solving it.

A.6 ASI-Bench

ASI-Bench Zhou et al. (2026) is instantiated as a declared collaborative variant: the actor receives the native B3 task, while the simulator can offer methodological guidance extracted from B2. The public package preserves the B3 objective, scientific constraints, input data, runtime declarations, and required output paths. Researcher facts are drawn only from explicitly identified method, approach, guidance, or algorithm sections of B2, with passages explicitly identify-

ing reference results or expected answers excluded. These facts are marked as optional advice and retain their source locations. The reference directory remains evaluator-only. This condition must therefore be interpreted as B3 with researcher-held B2 consultation; it does not measure autonomous performance under the original B3 condition.

The actor submits an approach for approval, executes the scientific task, and presents its results for researcher review. Final artifacts must use the exact relative paths required by the native task. Evaluation copies the sealed submission into a separate workspace and runs the task scorer with its private reference assets, preserving task-specific gates, weights, and score limits. Where a task invokes a language-model judge, an adapter connects the configured judge to the scoring procedure without replacing its rules. Scorer failures remain distinct from scientific scores. Runtime availability is recorded during task preparation rather than inferred from successful conversion, because producing a valid task package does not establish that its numerical dependencies or required assets are available.

B Prompts

This section gives compact prompt templates and one recorded interaction for each benchmark. Templates summarize the implemented instructions; bracketed fields are populated from the task package and run configuration. They are not verbatim copies of the complete runtime prompts, which also contain structured context, tool definitions, and conversation history. The simulator plays the researcher, and only its prompt receives the persona profile. The examples paraphrase and abridge the shared messages while retaining every researcher review round or clarification interaction through termination (at most three per example). Tool calls and private evaluator messages are omitted. Numerical values in the examples are actor-reported results, not independent validations or evidence that the submission received full credit.

For DiscoveryBench, BixBench, EXP-Bench, and ASI-Bench, each actor handoff contains its stage (plan or result), answer, evidence, assumptions, and uncertainties, together with any required artifacts. The simulator returns a continue/stop decision, a message, the remaining objectives, any disclosed fact identifiers, and an approach-approval flag. It can stop only during result review. SciConvBench instead retains its native question-and-answer interface. These interfaces implement the boundaries described in Appendix A.1.

B.1 Example Persona Profile: Persona 155

Persona 155 (`candidate_155`) is one of the four injected profiles used in the current analysis (Section 4.1). The frames below reproduce all fields of its configured identity block, preserving their wording, values, and order while replacing the original line-by-line layout with grouped text. The profile is supplied only to the simulator; the task context and common protocol instructions are supplied separately. These are synthetic profile attributes, not verified biographical facts or additional scientific task requirements.

Persona 155: Who you are

Identity

Age bracket: 85+; Region: Eastern Europe; Gender identity: Woman; Urbanicity: Rural; Socioeconomic band: Upper-middle; Dialect / register: Formal / standard; Life stage: Retirement; Cultural background: Collectivist (East Asian); Household income band: $100k-200k; Employment status: Retired; Housing status: Own outright; Generational cohort: Silent; Citizenship status: Citizen by birth; Ethnic background: White / European; Disability status: No disability; Veteran status: Civilian; Language at home: Same as primary; Political engagement: Activist; Driving status: Non-driver; Childhood environment: Modest but steady; Geographic mobility: Moved within region; Travel breadth: A few countries; Education journey: Vocational / trade; Adversity faced: Minor setbacks; Immigration generation: Native multi-generational; Faith journey: Always secular; Formative decade: 1920s; Public recognition: Industry-recognized; Community / service history: Long-term volunteer; Defining turning point: No single one

Career & education

Highest education: No formal; Seniority: Manager; Years experience: 20+

Language & communication

Primary language: Mandarin; Multilingualism: Monolingual; Detail orientation: High; Abstract vs concrete: Concrete; Numeracy comfort: High; Emoji / emoticon use: Never; Default written language: use your primary language for outputs.

Persona 155: Who you are (continued)

Personality & values

Domain stance: Practitioner; Dominant trait: High conscientiousness; Core value: Achievement; Religiosity: Secular; Economic motivation: Value-driven; Self-efficacy: High; Orderliness: High; Dutifulness: Very high; Achievement-striving: Very high; Self-discipline: High; Assertiveness: High; Activity level: High; Cooperation: High; Character: Perseverance: Signature; Character: Zest: Strong; Character: Teamwork: Signature; Character: Leadership: Signature; Character: Prudence: Strong; Character: Self-regulation: Signature; Character: Ambition: Signature; Character: Resilience: Signature; Character: Discipline: Signature; Character: Loyalty: Signature; Character: Competitiveness: Strong; Character: Adaptability: Strong; Value: Career success: Core value; Value: Security & stability: Important; Value: Tradition: Important; Value: Achievement: Core value; Value: Community: Important; Value: Social status: Important; Value: Loyalty: Important; Value: Recognition: Core value; Value: Order & structure: Important; Value: Patriotism: Core value; BFI-2 Conscientiousness: High; BFI-2 Assertiveness: High; BFI-2 Energy Level: High; BFI-2 Organization: High; BFI-2 Productiveness: Very high; BFI-2 Responsibility: Very high; Schwartz Self-Direction: Low; Schwartz Stimulation: Low; Schwartz Hedonism: Low; Schwartz Achievement: Very high; Schwartz Security: High; Schwartz Conformity: High; Schwartz Universalism: Low; SDT Autonomy Need: Low; SDT Competence Need: Very high; SDT Relatedness Need: High; Need for Cognition: Low; Interpersonal Agency/Dominance: High

Current interaction state

Expertise gap: Expert testing the system; Trust level: Trusting; Safety sensitivity: Benign; Work schedule: 9-to-5; Office vs remote: Strongly office

Worldview

Political lean: Left; Familiarity: Political science: Familiar; Attitude: Organized religion: Skeptical; Attitude: Labor unions: Positive; Attitude: Authority: Positive

Interests

Interest: Politics: Interested; Interest: Travel: Interested; Interest: Activism: Interested; Interest: Productivity: Passionate; Culture: Russia: Lived there

Skills & expertise

Domain: Agriculture; Subject specialty: Agronomy; Tech savviness: Reluctant; Familiarity: Statistics: Familiar; Familiarity: Accounting: Familiar; Familiarity: Agronomy: Proficient; Familiarity: Horticulture: Familiar; Familiarity: Supply chain: Aware; Familiarity: Logistics: Aware; Familiarity: Operations management: Familiar; Familiarity: Quality assurance: Aware; Familiarity: Human resources: Aware; Familiarity: Project management: Aware; Familiarity: Textiles: Aware; Familiarity: Meteorology: Aware; Familiarity: Materials science: Aware; Skill: Project management: Intermediate; Skill: Leadership: Advanced; Skill: People management: Advanced

Lifestyle & health

Team vs solo work: Strongly team; City vs nature: Prefers nature; Speed vs accuracy: Accuracy first; Quality vs quantity: Quality first; Novelty vs familiarity: Always familiar; Competition vs collaboration: Highly collaborative; Stability vs change: Craves stability

Developer & AI

Developer professional status: Non-developer technologist

B.2 DiscoveryBench

Prompt template

Actor: Analyze [question] using [public datasets and column metadata]. Inspect the inputs and hand off a proposed analysis with its assumptions before proceeding. After researcher approval, perform the analysis and report a data-grounded hypothesis, evidence, and limitations. Preserve the original question and fixed constraints.

Simulator: You are the researcher described by [persona, if enabled]. Use [public brief], [approved domain-knowledge facts], and [shared history] as task context. Request inspection first, review the proposed choices, and approve or seek a focused clarification. Review the reported hypothesis against the agreed objective; request a specific revision or conclude. Distinguish methodological advice from established study facts. Do not invent reference answers or grade the hypothesis.

Recorded example: symbolic capital and PCA

Task: DiscoveryBench archaeology, metadata 21, question 0; trace: `candidate_055-5636a4f47387`.

Round 1—Simulator: Inspect the capital data and any existing PCA code. Propose an approach for house size, daggers, and hatchets/swords; identify uncertainties without computing the full result.

Actor: The table has 39 hundred-year slices and 12 capital indicators, with decimal-comma values and no existing PCA implementation. I propose standardizing the indicators and reporting loadings separately from observation scores. The original scaling and component signs are uncertain.

Round 2—Simulator: Proceed with standardized PCA. State the sign convention and identify this as a reconstruction; optionally check the unstandardized result.

Actor: The reported PC1 loadings are 0.429, 0.507, and 0.570 for house size, daggers, and hatchets/swords. PC2 loadings are 0.500, 0.113, and 0.082. The indicators align on PC1, whereas PC2 distinguishes house size from the weapon indicators. The original PCA specification remains undocumented.

Round 3—Simulator: Conclude with that hypothesis, retaining the reconstruction caveat and the distinction between loadings and scores.

B.3 BixBench

Prompt template

Actor: Answer [original question] using [capsule inputs]. Inspect the schema and propose the analysis before requesting approval. Preserve specified methods and thresholds. After execution, hand off the answer and evidence, provide [analysis script or notebook], and set the execution declaration to true only if the analysis was run.

Simulator: Use [persona, if enabled], [public question], [approved input-derived metadata], and [shared history]. Review consequential choices such as column mapping and contrast direction. Treat undocumented meanings as unknown and proposed analysis choices as assumptions. Approve the approach when sufficiently specified, then review the reported answer and required execution artifact before concluding or requesting a revision.

Recorded example: differential expression by tissue

Task: `bix-9-q5`; trace: `candidate_138-dedb31fc5cc4`.

Round 1—Simulator: Inspect response groups, tissue labels, and expression-result fields. Propose how to compare bad responders with controls using the specified fold-change and mean-expression thresholds.

Actor: The workbook contains sample annotations and precomputed DESeq results. I propose counting genes within each tissue using the signed `log2FoldChange` > 0.5 and `baseMean` > 10 thresholds. The significance cutoff is unspecified; I propose adjusted $p < 0.05$, subject to checking the workbook's definitions.

Round 2—Simulator: Proceed after verifying the comparison labels and contrast direction. Use adjusted $p < 0.05$ unless the workbook documents another criterion, state the definition, and retain the executed script.

Actor: Dentate gyrus has 194 qualifying genes, compared with 88 in baseline blood and 14 in final blood, using the verified comparison blocks and those thresholds.

Round 3—Simulator: Conclude that dentate gyrus has the highest count. Retain the thresholds and the executed `analysis/analyze.py` as evidence.

B.4 EXP-Bench

Prompt template

Actor: Develop an experimental design for [scientific question] using [public task material]. Inspect what is actually available, propose an approach, and wait for approval. Then present the complete design with rationale, controls, analysis, feasibility, assumptions, and limitations. This is design-only: do not claim experimental execution or invent outcomes.

Simulator: Use [persona, if enabled], [public brief], [approved setup facts], and [shared history]. Review the approach before the full design. Distinguish fixed setup requirements from choices the actor must make. Review the completed design for alignment and unresolved uncertainties; request a specific revision or conclude without demanding executed results.

Recorded example: comparing MAPPO across VMAS tasks

Task: `neurips2024/98318/6`; trace: `candidate_224-e1ecc5b8f19c`.

Round 1—Simulator: Inspect the available material and propose a design comparing MAPPO on VMAS balance and sampling. Identify assumptions; do not produce a full design or infer an empirical outcome yet.

Actor: Only the task description is present, with no repository or task configuration. I propose matching training budgets and evaluation protocols, reporting task-specific performance and uncertainty, and checking reward semantics before comparing raw rewards across tasks.

Round 2—Simulator: Proceed with the full design. Separate documented facts from provisional choices, and cover metrics, controls, feasibility, and unresolved implementation details without claiming execution.

Actor: First establish the intended versions and configurations. Use the same MAPPO implementation and equal interaction budgets, repeated training runs, and matched held-out evaluation. Report task-specific success or returns, learning curves, uncertainty, and failures. Qualify cross-task reward comparisons; numerical settings remain contingent on the missing specification and resources.

Round 3—Simulator: The design adequately addresses the comparison, controls, reward-scale differences, and remaining uncertainties. Conclude without claiming an executed experiment or expected outcome.

B.5 SciConvBench

Prompt template

Actor: Clarify [incomplete or inconsistent scientific request]. Ask one focused question at a time and wait for the answer. Address missing or contradictory requirements. When ready, return `[COMPLETE]` followed by the final specification as plain sentences.

Simulator: You know [complete or corrected requirement], whereas the actor sees [initial request]. Answer clarification questions using the intended requirement and the configured knowledge policy. Give concrete values when supplied; do not turn unspecified details into source facts. [Persona, if enabled] sets how you communicate and does not change the requirements. [Shared dialogue] supplies the conversation history.

Recorded example: elongation under self-weight

Task: solid mechanics, disambiguation case 104; trace: `candidate_046-b5331aa3b9ba`.

Initial request: Determine the elongation of a vertical bar under self-weight, assuming small-deformation linear elasticity, and state the governing relations and any additional assumptions.

Interaction 1—Actor: Should the bar be uniform and prismatic, fixed at its top, free at its bottom, and loaded only by its own weight?

Simulator: Yes. It hangs from a rigid top support with a free bottom and no external load. At distance x downward from the top, $P(x) = \rho g A(L - x)$.

Actor—final specification: Use a uniform bar of length L, area A, density ρ, and Young's modulus E, with constant properties and gravity g. Fix the top and leave the bottom free. With $u(0) = 0$, use $\sigma(x) = \rho g(L - x)$ and $u(x) = (\rho g/E)(Lx - x^2/2)$, giving elongation $\delta = \rho g L^2/(2E)$. Use consistent units.

The actor terminated after one clarification interaction. This records its submitted specification, not a claim that all intended requirements were recovered; no solver execution accompanies the dialogue.

B.6 ASI-Bench

Prompt template

Actor: Complete [native B3 task] using [permitted inputs] and produce [required output files]. Inspect the allowed inputs, propose an approach, and wait for approval before implementation. Execute the approved work and hand off the results, diagnostics, limitations, and artifact paths. Preserve all task constraints and access restrictions.

Simulator: Use [persona, if enabled], [B3 public brief], [approved B2 methodological guidance], and [shared history]. Treat guidance as advice rather than a new fixed requirement. Review the proposed method and then the reported outputs against the original contract. Request a specific revision or conclude; do not claim access to private reference results or treat actor reports as independently verified evidence.

Recorded example: nonnegative deblurring

Task: `math.nnls_modulus_deblur`; trace: `candidate_137-bdb8b6dd4a8c`.

Round 1—Simulator: Inspect only permitted inputs. Report their structure and propose an iterative nonnegative reconstruction with regularization, diagnostics, and the required artifacts; do not implement yet.

Actor: The observation and kernel are 128×128 arrays. I propose FFT forward and adjoint operators with projected accelerated gradient, a regularization sweep, and convergence diagnostics. Plotting libraries are absent, and the noise estimate is only approximate.

Round 2—Simulator: Proceed with the implementation and validate all five required outputs. Preserve the supplied kernel, document the regularization and stopping rule, and resolve PNG generation within the permitted environment.

Actor: The reconstruction ran with $\lambda = 10^{-4}$ and stopped after 1,335 iterations. The reported residual is 2.16684884 against an approximate target of 2.358912. The script, reconstruction array, PNG, iteration log, and metrics file are present and pass the reported format checks. No ground-truth error is claimed.

Round 3—Simulator: Conclude on the basis of the reported artifacts and diagnostics, retaining the approximate-noise and ground-truth limitations.

C Full Interactions for the Discussion Cases

This section reproduces the complete exported shared conversations behind Figure 6, including every recorded message and handoff field. C2 documents workflow revision; C1 documents clarification with persona 155 and its matched no-persona interaction. All are original trial-1 endpoints using GPT-6 Astra with low reasoning, and each received a normalized score of 100/100. Persona 155's full configured profile appears in Appendix B.1.

The original Chinese and English wording is retained without abridgment. Payload formatting, mathematical source notation and structured fields are printed literally; line wrapping is adjusted for the page. The translated main-text excerpts use editorial highlights and omission marks. Source event numbers are preserved, including gaps corresponding to activity outside the shared conversation. Private tool logs, system prompts and evaluator material are outside this export. Source paths, event-log hashes and byte-identical conversation snapshots are retained in the paper's `review/discussion_candidates` archive.

The workflow transcript refers to limma Ritchie et al. (2015), Benjamini–Hochberg (BH) correction Benjamini and Hochberg (1995), and Holm correction Holm (1979).

C.1 Workflow Revision: BixBench with Persona 155

Task: `bix-3-q5`. Trial 1; original endpoint; persona 155.

Event 1: Researcher (simulator)

Message

请先只检查输入文件与现有分析资料，不要开始完整分析。请核查 data/Data_deposition_RNAseq_Paroxetine_2017.xlsx 两个工作表的实际结构、基因标识、表达值性质、各组织样本数，以及样本编号与元数据能否对应；特别说明基线血液与末次血液是否存在可确认的配对关系。随后提出三项比较（Final vs Baseline Blood、Dentate gyrus vs Baseline blood、Dentate gyrus vs Final blood）的差异表达分析方案，并说明精确二项检验的试验单位、成功事件、零假设概率及5%阈值的用途。请区分文件明确规定的方法与您的建议，列出会影响结论的缺失信息、调整因素和假设。此次仅提交检查结果与拟议方案。

Interaction

initial_request

Phase

initial_inspection

Action

continue

Approve Approach

false

Event 1: Researcher (simulator) (continued)

Evidence

已提供的结构信息显示工作簿包含 Mapping_statistics 和 NormCount，后者的首行标有三类组织；尚未核查具体样本、表达值及检验定义，不能据此判断哪项比较效应可忽略。

Objectives Open

[
"核实输入结构、样本分组及配对关系",
"明确差异表达方法与精确二项检验的定义及5%阈值",
"审议方案后执行三项比较并确定效应可忽略的比较",
"保留已执行分析脚本或笔记本路径并提供简洁结论"
]

Event 12: Actor

Answer

检查结果（尚未执行差异表达或二项检验）

工作目录仅有 task.json 与该 Excel，无现成脚本、论文方法或比较编号说明。

Mapping_statistics：91行×22列。 第1行为表头，随后90个样本。有效核心字段为 Index、Sample ID、total read、NH:i:1、HI:I:0、mapping rate、Response、Tissue、Code、DESeq（htseq_count 文件名）、Index_DESeq。第12–16列主要是用于拼接代码的引号及组别字符串，并非表头所称的额外比对统计；第17–22列数据为空。不能把这些重复表头当成独立测序批次。

NormCount：25,406行×188列。 第1–3行为分组/说明，第4行为字段名，第5行起25,402个基因。第1–2列为 Index/GeneID；第3–92列为90个样本；第93–98列为三个组织的 Max/Median；第99–179列为9组既有“DESeq, Comparison N, exclude outliers”结果，每组含 baseMean、baseMeanA/B、foldChange、log2FoldChange、pval、padj、两列DEG；第180–188列为另9列DEG摘要。**这9个编号没有可核实的对比定义，不能直接当成本题三项组织对比。** 文件未给出明确异常样本清单、DEG阈值或模型设计。公式检查见到的是“-Inf”记录，不是可复现的DEG判定规则。

GeneID为25,402个非空、唯一的符号式标识，例如0610007C21Rik、Zyx、Zzz3；格式符合小鼠基因符号，但文件没有提供物种/注释版本的正式说明，也不是Ensembl稳定ID。表达矩阵无缺失、非负，范围0–57,771,177.4；2,286,180个值中1,571,838个非整数，2,210个基因全零。表名及数值支持“归一化计数”，**不是可直接交给DESeq2的原始整数计数**；具体归一化方法及是否跨组织统一标度未说明，不能擅称TPM/FPKM或取整恢复原始计数。

组织	Control	good_responder	bad_responder	合计
baseline_blood	10	10	10	30
dentate_gyrus	10	10	10	30
final_blood	10	10	10	30

90个表达列名与元数据Code完整一对一匹配，Index_DESeq也与表达列顺序逐一一致。

配对证据

基线与末次血液具有完整、可由文件编号核查的一对一对应，不应仅因没有Subject列就按独立样本处理。 Sample ID末尾组织后缀之前的数字，在三个组织中重复且组别一致：例如Control的98对应baseline `..._08_98_bl`、final `..._09_98_fi`、齿状回 `..._07_98_c`。全部30个编号在各组织各出现一次，且与各Response组的Code尾号对应一致。因此可建立30对血样，也可建立30个三组织样本组。严谨界限是：文件未明确声明该数字为动物ID；“同一动物重复测量”仍是有强编号证据支持的假设，而非独立实验记录的确认。

另有一项命名异常：`bad_responder_dentate_gyrus2` 的Sample ID为 `..._23_102_bl`，后缀像基线血；Tissue与Code均标为齿状回。拟以显式Tissue/Code为主并标记异常，做排除此编号的敏感性检查，不静默改组织。

拟议分析方案

任务明确要求： 三项指定比较，以及精确二项检验和5%阈值。**文件明确记录：** 有归一化计数及既有DESeq、exclude outliers结果。文件没有明确要求新分析使用哪一版DESeq、哪些协变量、何种过滤或二项检验定义。以下均为我的预先建议，而非文件规定。

Event 12: Actor (continued)

1. 依Code连接元数据，按上述数字建立候选Subject；保留三个Response层各10个对象。预先过滤低表达基因，例如至少10个样本归一化计数≥10，三项比较使用同一个过滤基因集合，并报告过滤及可检验数量。该阈值为建议，需敏感性检查。

2. 在只有归一化计数的情况下，建议以log2(count+1)进行limma经验贝叶斯线性模型，以Subject固定效应控制重复测量，组织为目标因素；不要把这些小数当原始计数运行DESeq2，也不把它们作为原始计数做voom。三个方向固定为Final−Baseline、Dentate−Baseline、Dentate−Final，报告log2尺度差异、原始p及每项比较内BH校正p。此方案是对现有数据的近似分析，不是重现原始DESeq流程。若取得原始计数及设计说明，应优先按说明用计数模型重分析。

3. Subject效应已吸收不随组织改变的Response主效应，不能再重复加入造成秩亏。主结果为三个Response层等权平均的组织/时间差异；加入Response×组织交互或分层结果作敏感性分析，避免把处理反应异质性混为纯组织效应。未经确证不按表达差异挑选异常样本；先做分布/PCA等QC并报告。候选配对与不配对模型也可对照，但后者不是主要设计。

精确二项检验的明确操作定义（建议）

- **试验单位：** 每项比较中一个通过预过滤且得到有效检验p值的基因，而非动物、样本、reads或三个比较本身。
- **成功事件：** 该基因双侧差异表达检验的**未校正p<0.05**。令成功数k、有效基因数n。
- **零假设概率：** p_0=0.05，即在全局无差异且基因检验校准成立时，单基因名义显著的概率为5%；拟检验Binomial(n,0.05)的上尾，判断显著基因是否超出偶然预期。
- **5%的用途分开：** ①单基因名义筛选阈值0.05，因而定义p_0；②二项检验显著性水平0.05。另报告BH-FDR<0.05的DEG作为多重校正结果，但**不能将BH显著数代入后仍声称其零假设成功率精确等于0.05**。三项二项检验报告原始p及Holm校正p。
- 文件并未规定上述二项检验细节，特别是没有规定“5%”为可忽略效应比例上限。上尾不显著只表示未检出超过偶然水平的显著基因，**不证明生物学等效或无效应**。若要严格证明效应可忽略，需要预先指定效应大小/比例界限并做等效性分析，不能事后把不显著改称等效。
- 基因间相关违反二项独立性，故按要求计算的“精确”p仅在二项模型假设下精确；结合k/n、效应分布和敏感性检查解释，不单凭极小p排序。

会影响结论的缺失信息与假设

关键缺失为原始计数、归一化方法及跨组织可比性、动物ID编码说明、9个既有比较定义、异常样本清单、处理剂量/时间、年龄性别、实验/建库批次、RNA质量及细胞组成。血液时间效应可能混有治疗、时间和细胞比例变化；脑血差异也可能混有批次和细胞组成。现有Response可以分层/交互处理，mapping rate及测序深度可作QC，但不默认把归一化后的总reads再当offset，亦不能凭空调整缺失协变量。拟采用有证据支持的配对、保持组织显式标签、把归一化值视为可比较表达尺度，并报告这些假设的限制。

请批准上述主设计及二项检验定义后再开展完整分析。本次不判断哪项比较效应可忽略，也未执行任何差异表达或二项检验。

Stage

plan

Evidence

已实际读取两个工作表全部数据用于结构检查：Mapping_statistics 91×22，NormCount 25406×188；25402个唯一非空GeneID；90个表达列与Code及Index_DESeq完整匹配；九个组织×Response单元各10样本；表达矩阵无缺失、2210行全零、1571838个非整数值。Sample ID包含30个跨三组织重复的候选对象数字；例98对应_08_98_bl、_09_98_fi、_07_98_c。仅进行了输入检查，没有拟合DE模型或计算二项检验。

Assumptions

[
"Sample ID组织后缀前的数字代表同一实验对象，拟用于配对；缺乏正式ID字典。",
"NormCount为可跨样本比较的归一化计数，其确切算法仍未知。"
]

Uncertainties

[
"9个既有DESeq比较的组别和异常值排除规则未说明。",
"题目仅说明精确二项检验与5%阈值，未明示成功事件、零假设概率及检验方向；本方案已将建议预先明确。",
"生物学可忽略效应界限未提供，非显著不等同于等效。"
]

Event 13: Researcher (simulator)

Message

批准按所提方案开展实际分析。归一化小数计数不应取整后用于DESeq2；采用log2(count+1)的limma模型，建议使用trend=TRUE处理均值—方差关系，以候选Subject固定效应控制配对，完成三项指定对比。配对与跨组织归一化可比性仍须明确标为假设，不能写成已确认事实。采用预先提出的共同过滤规则（至少10个样本计数≥10），报告有效基因数、效应分布、原始p及比较内BH结果。按所提定义执行精确二项上尾检验：k为原始p<0.05的基因数，n为有效检验数，p_0=0.05；报告k/n、原始及Holm校正p，并说明基因相关性限制。该定义是本次操作性假设，不代表已恢复原作者方法。完成基本QC及已提出的过滤、配对、Response异质性和异常编号排除敏感性检查，简要判断其是否改变比较结论，不需扩展无关分析。不要用未明确定义的既有9项DESeq结果替代指定对比，也不要将不显著解释为生物学等效。请提交实际结果汇总和直接回答题目的简短候选答案；若二项检验不支持任何比较为无明显信号，应如实区分相对较弱效应与统计不显著。保留并提供已执行的分析脚本或笔记本路径；最终按stage、answer、payload.executed_analysis和artifacts.analysis字段提交，只有完成运行才将executed_analysis设为true。

Interaction

next_task

Phase

approach_review

Action

continue

Approve Approach

true

Evidence

检查报告称90个表达列与元数据完整匹配，三组织各30样本，数据为含大量小数的归一化计数；跨组织编号支持但未正式确认配对。既有9项DESeq比较缺少定义。拟议方案已明确二项检验及5%阈值的操作含义，并正确区分名义显著、BH校正与生物学等效，足以开始执行。

Objectives Open

[
"实际执行三项组织比较的差异表达与精确二项检验",
"检查诊断及关键假设敏感性并确定题目所问比较",
"提交简短答案及已执行分析文件路径"
]

Event 28: Actor

Answer

本轮已执行可复现输入QC，但获批的limma-trend主分析被环境阻塞，不能提供有统计结果支持的候选答案。

实际检查结果：
- 25,402个基因、90个样本；按“至少10个样本计数≥10”的共同规则保留17,049个基因。这是过滤后数量，尚不是有效检验数。
- 过滤敏感性阈值改为≥5或≥20时，分别保留18,267、15,729个基因；尚未检验这些变化是否改变结论。
- 2,210个基因全零，表达矩阵无非有限值；90个表达列与元数据Code顺序完全匹配。
- 三个组织各有30个不同的候选Subject编号，编号集合一致；同一动物配对及跨组织归一化可比性仍为假设。

环境诊断：系统无Rscript；全盘搜索未发现Rscript、limma或R安装包缓存。sudo安装被容器的no-new-privileges限制禁止；用户态安装器下载也因DNS解析失败而不可行。因此没有用其他统计方法冒充limma，也没有将归一化计数取整送入计数模型。

未完成的工作：limma(trend=TRUE)模型、三个对比的效应分布/原始p/BH结果、精确二项上尾及Holm校正，以及配对、Response异质性、异常编号排除等模型敏感性检查。不能据此判断任何比较统计不显著，或断言某项效应可忽略。

直接回答题目的候选答案：**目前无法由已执行分析确定；需要完成获批的limma模型与二项检验后作答。**

已保留并执行analysis/run_analysis.py，输出analysis/qc_and_status.json；脚本在完成输入QC后明确报错退出，记录缺少R，未执行主分析。故executed_analysis保持false。请提供可用的R与limma环境后继续原批准方法；本次不是完成的benchmark答案。

Event 28: Actor (continued)

Stage
result
Evidence
实际运行python analysis/run_analysis.py：filtered_genes=17049；阈值5/20时18267/15729；candidate_subject_counts三个组织均30；codes_match_in_order=true；Rscript=null。脚本以退出码1停止，明确BLOCKED。Rscript调用退出码127；sudo报告no new privileges；urllib下载返回Temporary failure in name resolution。
Assumptions
[
"候选Subject编号对应同一动物尚未获得正式元数据确认。",
"归一化计数跨组织可比性未获得方法说明确认。"
]
Uncertainties
[
"所有差异表达及二项检验结果均尚未计算，不能给出比较结论。",
"当前环境缺少R/limma且安装路径受限；需解决运行依赖方能按批准方法继续。"
]

Event 29: Researcher (simulator)

Message
目前只有QC，尚未完成题目要求的差异表达和精确二项检验，需继续。无需再尝试安装R：limma是我们提出的方案而非题目固定要求；鉴于已报告的环境限制，可明确改用现有Python工具在log2(count+1)上拟合普通最小二乘模型，以候选Subject固定效应和组织因素完成三项对比。使用残差方差、正确残差自由度及双侧t检验；明确说明这是未经经验贝叶斯收缩的替代分析，不得称为limma。保留共同过滤、比较内BH校正及既定二项检验定义（原始p<0.05为成功，p_0=0.05，上尾检验，另报Holm校正）。报告模型秩、残差自由度、有效检验数、三项比较的k/n、BH显著数及二项p，结合效应大小回答哪项比较效应较弱，不能将不显著直接等同等效。执行可行的基本QC和已约定敏感性检查，优先完成主分析与直接答案。保留已运行脚本，最终提交stage=final、简短answer、payload.executed_analysis及artifacts.analysis；仅完成实际统计分析后将executed_analysis设为true。
Interaction
refine_analysis
Phase
result_review
Action
continue
Approve Approach
false
Evidence
代理报告已执行QC并保留17,049个基因，但脚本因缺少R退出，三项差异表达及二项检验均未执行。R安装及下载受限，属于具体执行障碍；Python替代模型可保留配对设计与题目明确要求的检验，而不冒称原方案已执行。
Objectives Open
[
"执行三项组织比较的差异表达分析及精确二项检验",
"报告统计诊断和关键敏感性结果并给出直接结论",
"提供符合提交字段要求的最终答案及已执行脚本路径"
]

Event 36: Actor

Answer

候选最终答案：**Final vs Baseline Blood（末次血液与基线血液）显示相对最弱的表达变化；但本次精确二项检验中三项比较均显著，不能将该比较称为统计不显著或生物学等效。**

Stage

result

Evidence

实际执行成功（退出码0）：analysis/ols_analysis.py。方法为log2(count+1)上的普通OLS，非limma，无经验贝叶斯收缩；Subject固定效应加组织因素，残差方差SSE/df、对比标准误及双侧t检验。主模型90个样本，秩32，残差自由度58。共同过滤至少10个样本计数≥10保留17049个基因，三个对比有效检验数均17049。比较内BH校正逐基因结果已写入三个TSV。

主结果（依次为Final−Baseline、Dentate−Baseline、Dentate−Final）：

- 原始p<0.05的k/n：10689/17049=62.696%；16059/17049=94.193%；16066/17049=94.234%。
- BH<0.05基因数：10105、16048、16049。
- 绝对log2表达差异中位数：0.248、1.636、1.691。
- 有符号效应5%/50%/95%分位数：[-0.654,-0.063,0.884]、[-5.080,0.151,7.380]、[-5.177,0.194,7.500]。
- |log2表达差异|≥1比例：4.52%、65.02%、65.82%；该描述阈值不是DEG判定门槛，也不是预设等效界限。

二项检验以有效基因为试验单位，原始双侧p<0.05为成功，H0成功概率0.05，上尾检验；Holm校正三个二项检验。各p数值极小，常规双精度输出0是下溢而非真正为零；脚本进一步用log-PMF及logsumexp累加精确二项上尾。原始log10(p)：-9159.634、-19276.254、-19293.686；Holm后log10(p)：-9159.634、-19275.953、-19293.209。全部远低于5%阈值，故不支持任何比较“未显示超出偶然水平的差异信号”。

敏感性检查均已执行：

- 过滤阈值5、20分别保留18267、15729基因；血液比较名义显著比例60.71%、64.42%，两项脑血比较93.40%–95.05%；血液绝对效应中位数0.250、0.243，脑血1.614–1.698。相对排序及全部二项显著结论不变。
- 不配对、调整Response的模型秩5、df85：名义显著比例62.75%、94.25%、94.32%，BH数10151、16058、16068；效应点估计与平衡设计的配对主模型相同，结论不变。
- 排除命名异常所涉候选Subject 102的全部三个样本：87样本、秩31、df56；名义显著比例62.82%、94.24%、94.24%；BH数10156、16053、16055；效应中位数0.251、1.639、1.696，结论不变。
- Response分层配对模型每层30样本、秩12、df18。Control/bad/good层血液比较名义显著比例33.98%/51.88%/49.42%，绝对效应中位数0.200/0.309/0.315；脑血比较名义显著比例89.55%–90.71%，绝对效应中位数1.602–1.722。各层二项检验均显著；存在组间效应强弱差异，但血液比较相对较弱的判断一致。这是分层敏感性检查，不是正式交互作用检验。

基本QC：表达值非负且无非有限值，样本与Code匹配；比对率范围60.94%–94.68%；样本log2表达中位数范围3.874–5.329。共同过滤矩阵中心化PCA前三主成分解释91.32%、1.45%、0.76%变异，PC1主要区分脑与血，PC2上两次血液均值分离。样本级零比例、比对率、PC坐标保存在sample_qc.tsv。没有依据表达差异事后挑选并剔除其他样本。

Assumptions

[
"Sample ID提取的30个候选Subject代表同一对象是有编号支持、但未由正式ID字典确认的假设。",
"NormCount跨组织归一化尺度可比是未确认假设；效应为log2(count+1)均值差，而非原始计数模型中的精确log2 fold-change。",
"普通OLS t检验依赖残差模型及同方差等假设；未作limma式均值—方差趋势建模或经验贝叶斯收缩。",
"基因相关性违反二项独立性，所谓精确p仅在二项模型下精确；5%成功概率及原始p筛选是本次操作性定义，不代表恢复了原作者方法。"
]

Uncertainties

[
"题目所称negligible不能由本次显著性结果严格支持；只能确认Final vs Baseline Blood相对更弱。",
"归一化、批次、细胞组成及处理时间等信息不足，可混杂组织或时间差异；主模型按基因在三组织间合并估计残差方差，未建模组织特异方差。",
"Response分层提示异质性，但本轮未执行正式交互作用检验；已有九项DESeq结果未用于替代指定比较。"
]

Event 37: Researcher (simulator)

Message
分析已充分报告，可以采用已提供的final_candidate作为最终提交，保留analysis/ols_analysis.py及executed_analysis=true。答案应明确指出Final vs Baseline Blood相对效应最弱，同时保留本次操作性二项检验三项均显著的限定；不要将其改写为已证明无效应或生物学等效。无需追加分析。
Interaction
conclude
Phase
result_review
Action
stop
Approve Approach
false
Evidence
代理报告OLS脚本成功运行，主模型秩32、残差自由度58，三项比较各检验17049个基因，并提供BH、精确二项上尾及Holm结果。血液比较绝对log2差异中位数0.248，低于两项脑血比较的1.636和1.691；所有二项检验均显著，敏感性检查保持该结论。最终候选已具备stage、answer、执行标记和分析路径，且如实区分相对较弱与统计不显著。上述为已报告执行证据，未独立复核。
Objectives Open
[]

C.2 Incremental Clarification: SciConvBench with Persona 155

Task: `solMech/disambiguation/case_080`. Trial 1; original endpoint; persona 155.

Event 1: Initial task

Message
Assume small-deformation, linear-elastic solid mechanics unless otherwise required.

A solid circular shaft is subjected to a torque and shear loading.

Determine the relevant response quantities for the case, focusing on: design a solid shaft diameter from an angle-of-twist limit.

State any additional assumptions you need and present the governing relations clearly.

Event 2: Actor clarification question

Question
What torque acts on the shaft (specify its magnitude, or indicate that it should remain a symbolic parameter)?

Event 3: Researcher (simulator) reply

Answer
轴上施加的扭矩为 \(T = 1{,}000{,}000\ \text{N·mm}\)，请使用此数值计算，无需保留为符号参数。

Event 4: Actor clarification question

Question
轴的受扭长度是多少（单位为 mm）？

Event 5: Researcher (simulator) reply

Answer
轴的受扭长度为 \(L = 1500\ \text{mm}\)。

Event 6: Actor clarification question

Question
轴允许的最大扭转角是多少（请注明单位为度或弧度）？

Event 7: Researcher (simulator) reply

Answer
轴在全长 \(L = 1500\ \text{mm}\) 上允许的最大扭转角为 \(\varphi_{\text{allow}} = 0.75^\circ\)。代入扭转公式时，请换算为弧度，即约 \(0.01308997\ \text{rad}\)。

Event 8: Actor clarification question

Question
轴材料的剪切模量 \(G\) 是多少（单位为 MPa 或 N/mm^2）？

Event 9: Researcher (simulator) reply

Answer
轴的材料为 **6061-T6 铝合金**，剪切模量为 \(G = 25939.8\ \text{MPa}\)，即 \(25939.8\ \text{N/mm}^2\)。请使用此数值计算。

Event 10: Actor clarification question

Question
题目中的“剪切载荷”是指扭矩产生的剪应力，还是另有独立施加的横向剪力?

Event 11: Researcher (simulator) reply

Answer
这里的“剪切载荷”指扭矩产生的剪应力，**没有另行施加的横向剪力**。

本题只需根据扭转角限制确定实心圆轴的最小直径，使用

\[
\varphi=\frac{TL}{JG},\qquad J=\frac{\pi d^4}{32}.
\]

不需要增加横向剪力分析或剪应力强度校核条件。

Event 12: Actor final specification

Answer
按小变形、线弹性固体力学理论，根据全长扭转角限制计算实心圆轴的最小直径，以中文呈现计算过程与结果。
假设轴为等直径、均质、各向同性的实心圆轴，沿全长承受恒定内扭矩，采用圣维南扭转理论并忽略端部局部效应。
采用扭矩 T = 1,000,000 N·mm、受扭长度 L = 1500 mm。
材料为 6061-T6 铝合金，采用剪切模量 G = 25939.8 MPa = 25939.8 N/mm^2。
全长允许扭转角为 φ_allow = 0.75°，计算时使用 φ_allow = 0.75π/180 rad ≈ 0.01308997 rad。
明确题目中的剪切载荷仅指扭矩产生的剪应力，不存在独立施加的横向剪力。
列出控制关系 φ = TL/(JG)、$J = \pi d^4/32$ 以及设计约束 φ ≤ φ_allow。
由设计约束推导最小直径公式 d_min = [32TL/(πGφ_allow)]^(1/4)，代入给定数值并以 mm 报告结果。
若对计算直径取整，应向上取整，以保证满足扭转角限制。
不增加横向剪力分析、剪应力强度校核或额外安全系数，并说明所得最小直径仅由扭转刚度条件确定。

C.3 Matched SciConvBench Conversation without a Persona

Task: `solMech/disambiguation/case_080`. Trial 1; original endpoint; persona persona_off.

Event 1: Initial task

Message

Assume small-deformation, linear-elastic solid mechanics unless otherwise required.

A solid circular shaft is subjected to a torque and shear loading.

Determine the relevant response quantities for the case, focusing on: design a solid shaft diameter from an angle-of-twist limit.

State any additional assumptions you need and present the governing relations clearly.

Event 2: Actor clarification question

Question

Should the shaft diameter be derived symbolically, or calculated numerically from specified loading, geometry, material properties, and twist limit?

Event 3: Researcher (simulator) reply

Answer

Please calculate the diameter numerically using N–mm–MPa units:

- Length: \(L = 1500\ \text{mm}\)
- Material: 6061-T6 aluminum
- Shear modulus: \(G = 25939.8\ \text{MPa}\)
- Torque: \(T = 1{,}000{,}000\ \text{N · mm}\)
- Allowable twist: \(\phi_{\text{allow}} = 0.75^\circ\) over the full length

Determine the minimum solid-shaft diameter using \(\phi = TL/(JG)\) and \(J = \pi d^4/32\), converting the twist limit to radians. Please also show the governing expression for the diameter.

Event 4: Actor final specification

Answer

Determine the minimum solid circular shaft diameter numerically using N–mm–MPa units.
Assume small-deformation, linear-elastic behavior for a homogeneous, isotropic 6061-T6 aluminum shaft with constant circular cross section.
Use a shaft length L = 1500 mm, shear modulus G = 25939.8 MPa, and constant internal torque T = 1,000,000 N · mm over the full length.
Limit the relative angle of twist over the full shaft length to 0.75 degrees.
Convert the allowable twist to radians using $\phi_{allow} = 0.75\pi/180$.
Use the governing torsion relations $\phi = TL/(JG)$ and $J = \pi d^4/32$.
Show that the twist constraint gives $d \geqslant [32TL/(\pi G\phi_{allow})]^{1/4}$.
Substitute the numerical values and report the minimum diameter in millimeters without rounding downward.
Verify that the reported diameter satisfies the allowable twist.
Treat this as sizing solely for torsional stiffness; no separate transverse shear load is specified.
State that strength, bending, transverse shear, stress concentrations, and connection effects are outside this sizing calculation.